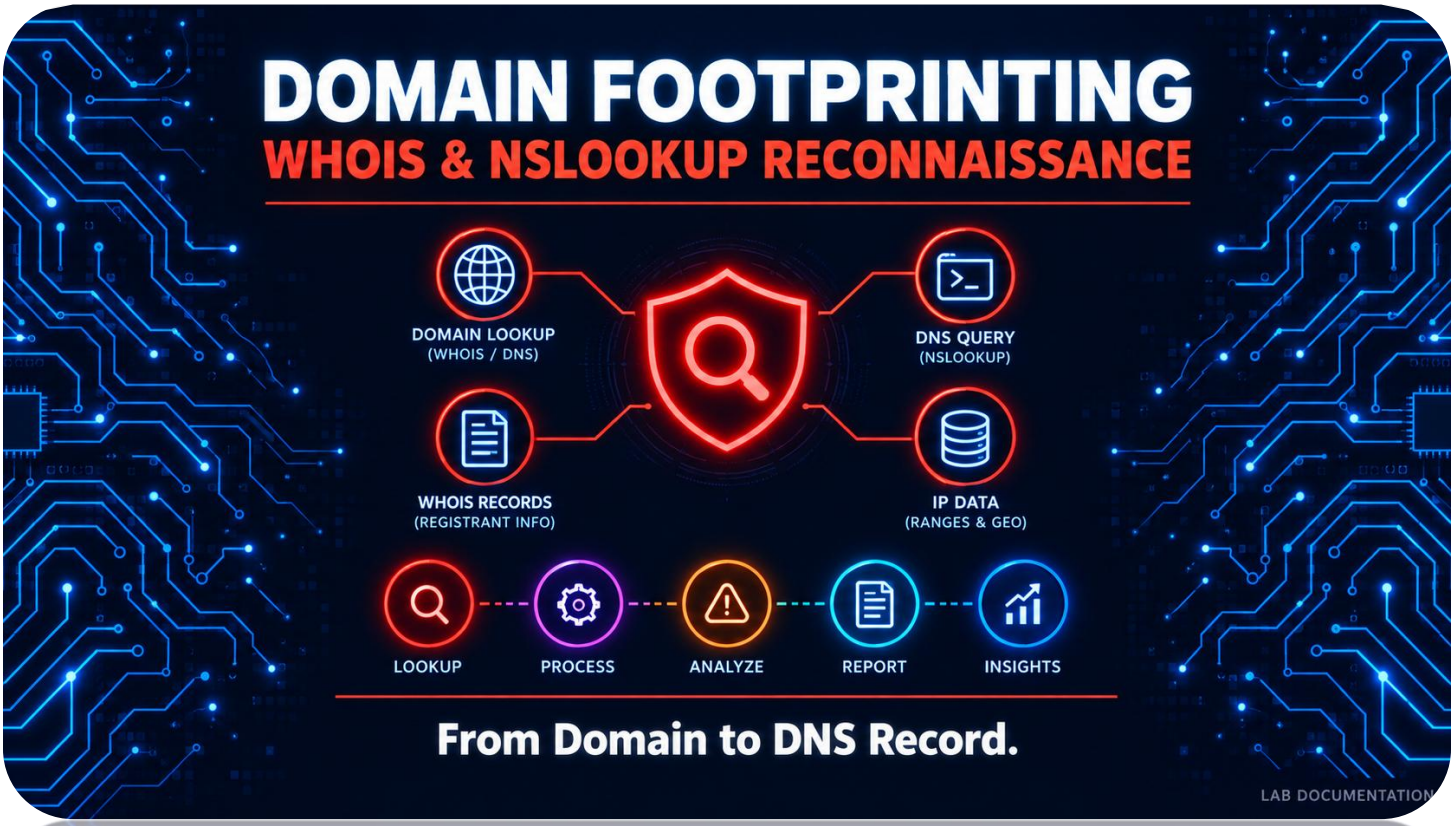


Abdul Wahid

Lab Report

Domain Footprinting – WHOIS & NSLookup

WHOIS

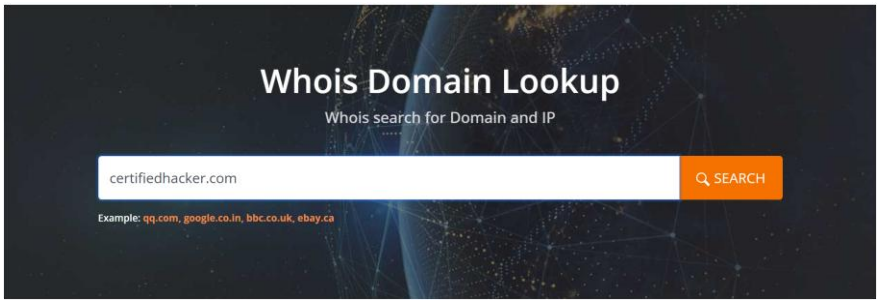
WHOIS is a query and response protocol used to search databases that store registered users or assignees of internet resources like domain names, IP addresses, and autonomous systems. It provides detailed information about domain ownership, registration dates, contact details, nameservers, and more.

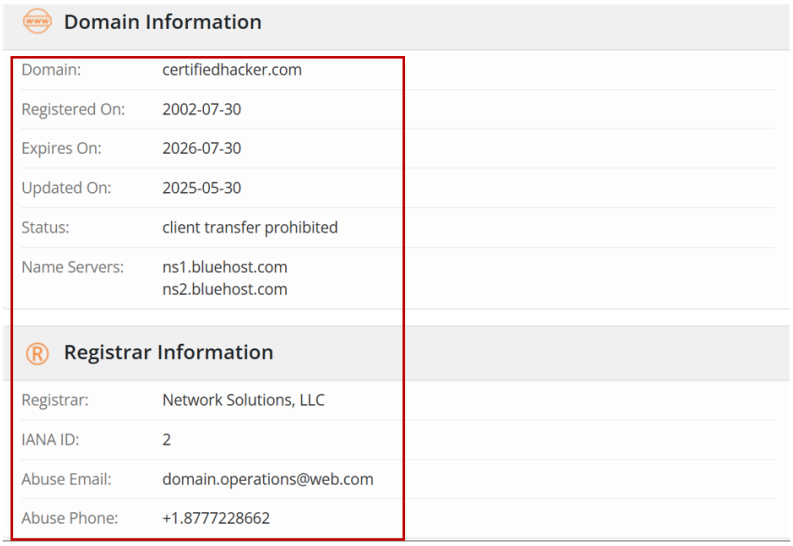
WHOIS Lookup

WHOIS Lookup is the process of querying WHOIS databases to retrieve registration and ownership information about a specific domain name or IP address. It's performed using command-line tools or web-based services to gather publicly available data about internet resources.

Two Ways to Perform WHOIS Lookup:

1. **Command Line:** Use terminal/command prompt with whois command
2. **Web-Based Tools:** Use online WHOIS lookup services for easy access





Example: Web-based WHOIS Lookup Tool - You can search domains and IPs directly from your browser

Popular Web-Based WHOIS Services:

- **who.is** - Simple and fast domain lookups
- **whois.com** - Comprehensive domain information
- **ICANN Lookup** - Official domain registration data
- **DomainTools** - Advanced domain research platform
- **Network Solutions** - Domain search and registration info

Why We Use WHOIS?

WHOIS is essential for various purposes including:

Security Analysis: Identify malicious domains and investigate cyber threats
Network Troubleshooting: Diagnose connectivity issues and identify network administrators
Domain Research: Check domain availability and investigate competitors
Legal Compliance: Verify ownership for legal disputes and trademark issues
Business Intelligence: Research potential partners or customers

❁ WHOIS Command Functions

Option	Full Name	Purpose	Example
-h	**Host**	Queries a specific WHOIS server instead of the default one	whois -h whois.arin.net 8.8.8.8
-p	**Port**	Connects to WHOIS server on a custom port number (default is 43)	whois -p 43 domain.com
-H	**Hide disclaimer**	Removes legal disclaimers and terms from output for cleaner results	whois -H google.com
-i	**Inverse lookup**	Performs reverse search to find domains owned by specific registrant or admin	whois -i admin-c ABC123
-T	**Type**	Specifies template query type for structured searches (person, domain, etc.)	whois -T person "John"
-K	**Connected**	Shows related records and associated domains with the queried resource	whois -K domain.com
-B	**Bulk**	Bypasses rate limiting for multiple queries (requires authorization)	whois -B domain.com
-r	**Raw**	Returns unformatted, unprocessed data directly from WHOIS server	whois -r domain.com
-v	**Verbose**	Displays detailed output including connection details and query process	whois -v domain.com
-V	**Version**	Shows the installed WHOIS client version and build information	whois -V

📋 Practical Examples

Example 1: Complete Domain Investigation

```
whois example.com | tee domain_full.txt | \
grep -E "Registrar|Email|Name Server|Creation|Expiry" > domain_summary.txt
```

Purpose: Saves complete output while extracting key information

Example 2: Email Extraction and Validation

```
whois domain.com | grep -Eoi '[A-Z0-9._%+-]+@[A-Z0-9.-]+\.[A-Z]{2,}' | \
sort -u > contact_emails.txt
```

Purpose: Extracts all unique email addresses from WHOIS data

Example 3: Bulk Domain Research

```
for domain in google.com facebook.com twitter.com; do
  echo "=== $domain ===" >> bulk_research.txt
  whois $domain | grep -E "Registrar:|Creation" >> bulk_research.txt
  echo "" >> bulk_research.txt
done
```

Purpose: Research multiple domains and compile results in one file

NSlookup

nslookup (Name Server Lookup) is a command-line tool used to query DNS (Domain Name System) servers to obtain domain name or IP address mapping information. It helps troubleshoot DNS-related issues by retrieving records such as A, AAAA, MX, NS, and more, allowing users to verify domain resolution, check server configurations, or detect issues like DNS hijacking.

Why Use nslookup

nslookup is essential for diagnosing and resolving DNS issues, making it a critical tool for network administrators, security professionals, and developers. Common use cases include:

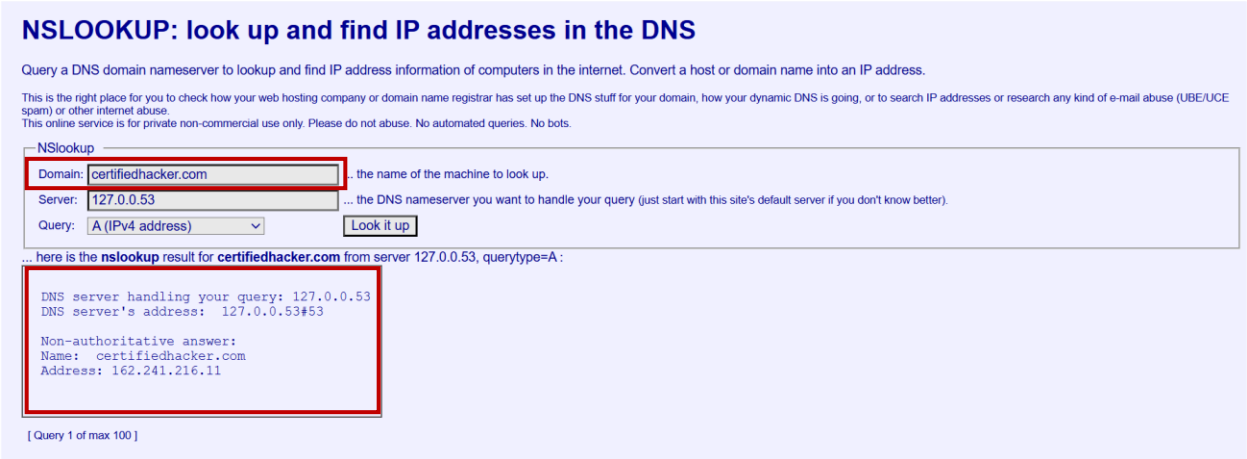
DNS Troubleshooting: Identify why a website or email service isn't resolving correctly
Security Analysis: Detect DNS hijacking or poisoning by verifying IP addresses
Domain Verification: Check domain records (e.g., MX, NS, TXT) for configuration accuracy
Network Diagnostics: Confirm DNS server functionality and response accuracy
Development & Testing: Validate DNS setups for new domains or services

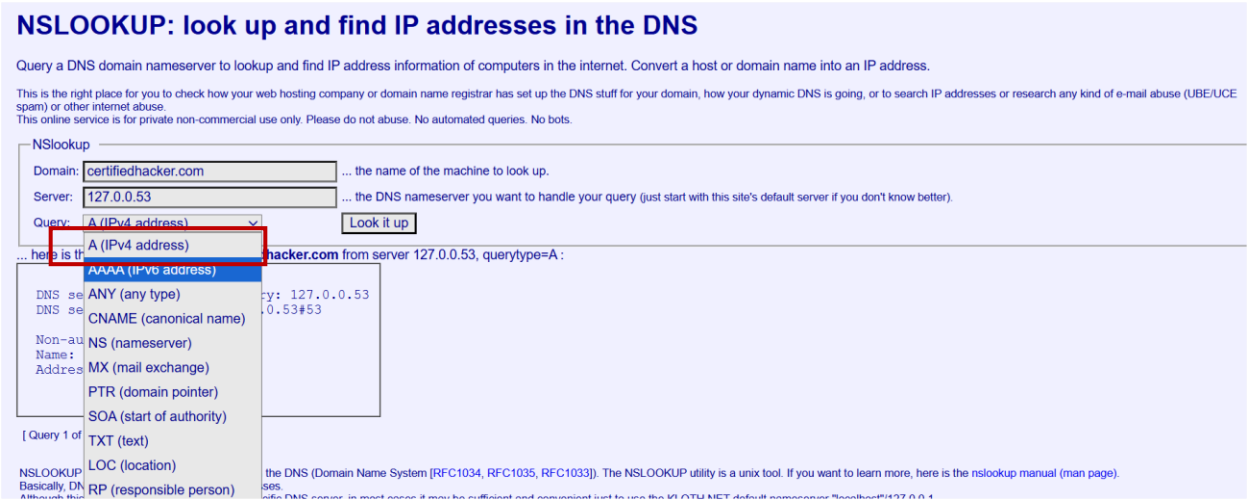
⚙️ How to Use nslookup

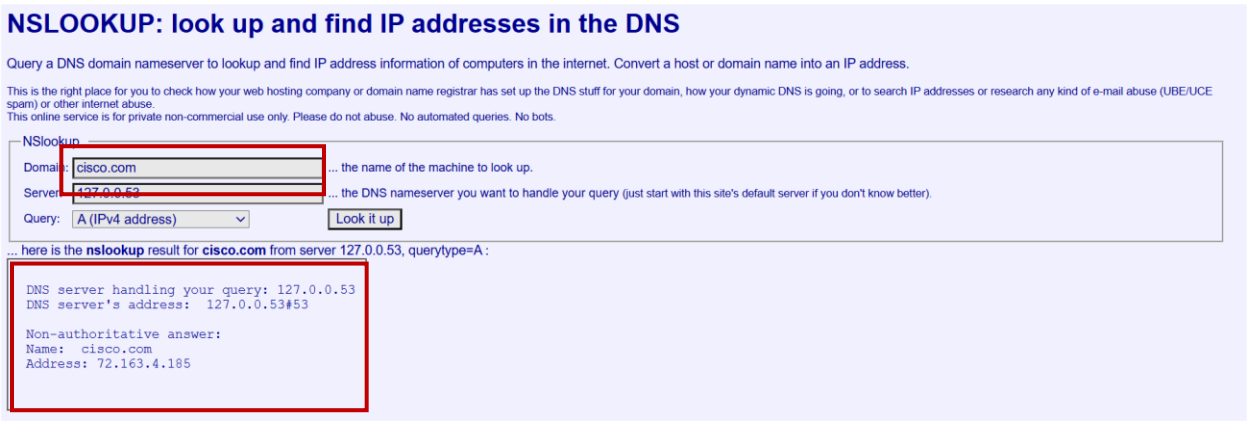
nslookup can be used in two modes: **interactive** (where you enter a command-line interface to run multiple queries) and **non-interactive** (single command queries). It allows querying specific DNS record types or custom DNS servers.

Two Ways to Use nslookup:

1. **Non-Interactive Mode:** Run a single query from the command line
2. **Interactive Mode:** Enter the nslookup shell for multiple queries







NSLOOKUP: look up and find IP addresses in the DNS

Query a DNS domain nameserver to lookup and find IP address information of computers in the internet. Convert a host or domain name into an IP address.

This is the right place for you to check how your web hosting company or domain name registrar has set up the DNS stuff for your domain, how your dynamic DNS is going, or to search IP addresses or research any kind of e-mail abuse (UBE/UCE spam) or other internet abuse.
This online service is for private non-commercial use only. Please do not abuse. No automated queries. No bots.

NSlookup

Domain: 8.8.8.8 ... the name of the machine to look up.

Server: localhost ... the DNS nameserver you want to handle your query (just start with this site's default server if you don't know better).

Query: A (IPv4 address) Look it up

... here is the **nslookup** result for **8.8.8.8** from server localhost, querytype=A :

```
;; Got SERVFAIL reply from ::1, trying next server
;; Got SERVFAIL reply from 127.0.0.1
DNS server handling your query: localhost
Address:       127.0.0.1#53

** server can't find 8.8.8.8.in-addr.arpa: SERVFAIL
```


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```
DNS server handling your query: localhost
DNS server's address:  ::1#53

Non-authoritative answer:
8.8.8.8.in-addr.arpa    name = dns.google.

Authoritative answers can be found from:
```

Basic nslookup Commands:

```
# Check A record for a domain
nslookup example.com

# Query a specific DNS server
nslookup example.com 8.8.8.8

# Check MX records
nslookup -type=MX example.com

# Enter interactive mode
nslookup
> set type=A
> example.com
```

Practical Example - Scenario

Problem: Website Not Opening

```
# Step 1: Check DNS resolution
nslookup example.com

# If error appears:
# "Server failed" or "Non-existent domain"
# Indicates DNS server issue

# Debugging Steps:
# 1. Check which DNS server is used
nslookup example.com
# Output: Server: 192.168.1.1 (Your router's DNS)

# 2. Try Google DNS
nslookup example.com 8.8.8.8
# If this works, local DNS has an issue

# 3. Check MX records (for email issues)
nslookup -type=MX gmail.com

# 4. Verify nameservers
nslookup -type=NS example.com

# 5. Check IPv6 support
nslookup -type=AAAA example.com
```

Common Errors & Solutions

1. "Connection timed out"

Cause: DNS server not responding or firewall blocking

Solution: Try different DNS server (8.8.8.8 or 1.1.1.1)

2. "Connection refused"

Cause: DNS server not accepting queries

Solution: Switch to public DNS server

3. "Non-authoritative answer"

Cause: Response from cache, not authoritative server

Solution: This is normal - cached responses are faster

4. "Server failed"

Cause: DNS server error or domain doesn't exist

Solution: Verify domain spelling and try another DNS server